

Data Cleaning, Transformation and Modelling Assignment Question-Theory

Question 1: What is Data Loading in Power BI and why is it considered the first step of analysis?

Data Loading in Microsoft Power BI refers to the process of importing data from different sources into Power BI so it can be cleaned, modeled, analyzed, and visualized. Data can be loaded from sources such as Excel files, SQL databases, cloud services, CSV files, websites, and APIs.

It is considered the **first step of analysis** because:

1. **Analysis cannot begin without data** – Power BI needs data to create reports and dashboards.
2. **Data quality affects results** – Incorrect or incomplete data leads to inaccurate insights.
3. **Foundation for modeling** – Relationships, calculations, and visualizations depend on properly loaded data.
4. **Supports decision-making** – Clean and structured data enables meaningful business analysis.

Example:

If a company wants to analyze sales performance, it must first load sales, customer, and product data into Power BI before creating charts or KPIs.

Question 2: Explain the difference between “Load” and “Transform Data” in Power BI.

Feature	Load	Transform Data
Meaning	Imports data directly into Power BI	Opens Power Query Editor to clean and modify data before loading
Purpose	Quick import of ready-to-use data	Prepare and improve data quality
Changes Allowed	No major editing	Filtering, merging, splitting, changing data types, removing nulls, etc.
Tool Used	Load option	Power Query Editor
Best Used When	Data is already clean	Data needs preprocessing

Example

- **Load:** Importing a clean Excel sales sheet directly into Power BI.
 - **Transform Data:** Removing duplicate customer records, changing date formats, or combining tables before analysis.
-

Question 3: What is a Fact Table and a Dimension Table? Give examples from the dataset.

Fact Table

A **Fact Table** contains measurable business data or numerical values used for analysis. It usually stores transactional records.

Characteristics:

- Contains metrics such as sales amount, quantity, profit, etc.
- Large number of rows
- Connected to dimension tables using keys

Example:

A **Sales** table containing:

- Order ID
- Product ID
- Customer ID
- Sales Amount
- Quantity
- Profit

This table is the **Fact Table** because it stores measurable business transactions.

Dimension Table

A **Dimension Table** contains descriptive information used to categorize and filter data in the fact table.

Characteristics:

- Contains textual or descriptive attributes
- Used for slicing and filtering reports
- Usually smaller than fact tables

Examples:

1. Customer Dimension

- Customer ID
- Customer Name
- City
- Segment

2. Product Dimension

- Product ID
- Product Name
- Category
- Subcategory

3. Date Dimension

- Date
- Month
- Quarter
- Year

Question 4: Why is Star Schema preferred over Snowflake Schema in Power BI?

In Microsoft Power BI, the **Star Schema** is generally preferred because it is simpler, faster, and more efficient for reporting and analysis.

Advantages of Star Schema

Advantage	Explanation
Simpler Structure	Fact table is directly connected to dimension tables, making the model easy to understand
Better Performance	Fewer joins improve query speed and report performance
Easy DAX Calculations	Relationships are straightforward, simplifying calculations
User-Friendly	Easier for report developers and business users
Optimized for BI Tools	Power BI works efficiently with star schema models

Snowflake Schema

A **Snowflake Schema** normalizes dimension tables into multiple related tables.

Example:

Instead of one Product table:

- Product → Category → Department

This reduces data redundancy but increases complexity and joins.

Why Star Schema is Better in Power BI

- Faster dashboard loading
- Easier relationship management
- Better filter propagation

- Simpler maintenance
- Improved report usability

Simple Comparison

Star Schema	Snowflake Schema
Denormalized	Normalized
Simple design	Complex design
Faster queries	More joins required
Preferred in Power BI	Less commonly used in Power BI